Inside an espresso machine: cold water to one shot

Shows how six parts turn cold water into a shot, in the order the water meets them.

STATUS	VERSION	DATE
exploration	v1.0	2026-10-29

1 Water tank

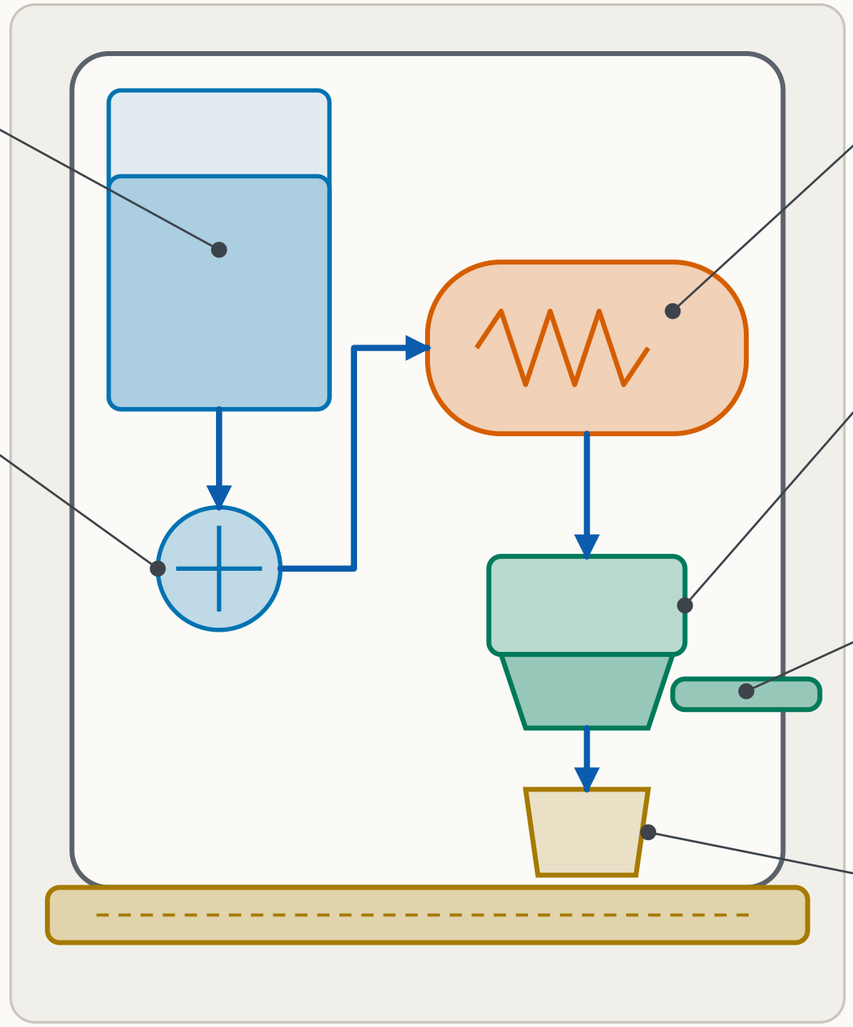
stores the cold water

about 2 L, refilled by hand

2 Pump

pushes water through the machine

about 9 bar of pressure



3 Boiler

heats the water

about 93 °C at the coffee

4 Group head

spreads water evenly over the coffee

about 25 to 30 s of flow

5 Portafilter and puck

holds 18 g of ground coffee

36 g of espresso comes out

6 Cup and drip tray

catches the shot and the spills

cup under the spout

ONE SHOT, IN NUMBERS

18 g in, 36 g out

in about 25 to 30 seconds

a common starting recipe, tune it to taste

LEGEND holds and moves water · heats · brews · catches · path of the water, in order 1 to 6

Specified: 9 bar, about 93 °C and an 18 g to 36 g shot in 25 to 30 s are common café starting values, not measurements of one machine. **Open:** real machines differ (heat exchanger, dual boiler).

Example page for the one-page-visual-design skill